

Lucas Nunes de Assis - Senior Python/Data Engineer

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Summary

Senior Python Engineer with 7+ years building backend services, REST APIs, and large-scale data platforms on Azure and AWS. Led a 6-person engineering team as acting team lead, designed real-time video analytics handling 200 concurrent streams from 100+ cameras, and built Medallion-architecture pipelines processing 500M+ rows with Spark, Databricks, and Airflow. Proven delivery across distributed international teams, from edge infrastructure and IoT messaging to LLM-powered agentic search.

Skills

- **Backend & APIs:** Python (Expert), FastAPI, Flask, REST APIs, Automated Testing (pytest, unittest, Selenium), Microservices, Real-Time Ingestion, OOP, Clean Code
- **Data Engineering:** ETL/ELT, Medallion Architecture, Apache Spark, PySpark, Databricks, Apache Airflow, Azure Data Factory, dbt, Snowflake, Google BigQuery, Data Modeling, Data Governance
- **Databases:** PostgreSQL, pgvector, MySQL, SQL Server, MongoDB, SQLite, Redis, Elasticsearch, Vector Databases
- **Cloud & DevOps:** Azure, AWS (EC2, ECS, Lambda, S3), Docker, Ansible, Terraform, Linux, Git, CI/CD, Infrastructure as Code
- **AI & Analytics:** Artificial Intelligence (AI), Machine Learning (Scikit-Learn, PyTorch, Pandas, NumPy), Computer Vision (YOLO, OpenCV), NLP, LLMs, Agentic AI, LangChain, Prompt Engineering, Power BI, Data Visualization
- **Streaming & IoT:** RTSP, WebRTC, MQTT, Edge Computing
- **Methodologies:** Agile (Scrum, Kanban), Project Management, Team Leadership, Mentoring

Languages: Portuguese (Native), English (Fluent), Spanish (Fluent)

Professional Experience

Paramount, Remote — Senior Python/Data Engineer

SEPTEMBER 2026 - PRESENT

Focuz Analytics, Remote — Senior Python/Data Engineer

MAY 2025 - JULY 2026

- Designed scalable Python/FastAPI backend services and edge services for an AI-first video analytics platform, handling 200 concurrent video streams from 100+ cameras across multiple distributed edge sites.
- Implemented natural-language video search using computer vision (YOLO, PyTorch) with PostgreSQL and pgvector, enabling users to find footage via plain-language queries instead of manually reviewing feeds.
- Built LLM-powered agentic AI workflows with LangChain, connecting language models to the platform's vector search and camera metadata for video analytics and search.
- Worked on the analytics data layer with dbt and Snowflake, structuring platform data for reporting and downstream analysis.
- Automated provisioning and management of edge sites with Ansible, and implemented MQTT-based messaging for IoT device control across the camera fleet, with RTSP/WebRTC live streaming (go2rtc).
- Containerized the full stack with Docker on Linux for reproducible deployments across edge and cloud environments.

Kantar IBOPE Media, São Paulo — Mid-Senior Python/Data Engineer

APRIL 2024 - MAY 2025

- Delivered end-to-end Python integrations with ad exchanges and the Meta Marketing API, producing audience-reach, impact, and conversion metrics segmented by demographics and device type (TV, mobile, social, streaming, VOD).
- Built and optimized large-scale Medallion-architecture data pipelines (ELT/ETL, ingestion, micro-aggregators) using Spark, Databricks, Airflow, and Azure Data Factory, processing 500M+ rows per project.
- Co-designed the refactoring of the platform's data system, improving pipeline performance and maintainability across the product.
- Delivered business-critical KPIs in Power BI on daily, periodic, and real-time cadences for cross-media campaign optimization used by global marketing teams.

- Became the team's reference point for connecting advertising business logic to technical implementation across multiple projects in a global team (LATAM, Europe, Asia).

BDO Brazil, São Paulo — Software Engineer

MAY 2021 - APRIL 2024

Progressed from Junior to Mid-Senior Engineer; served as Acting Team Lead for the final 9 months.

- Led a 6-person engineering team for approximately one year after the manager's departure: owned task management and prioritization, and acted as the primary interface between company Partners (Product Owners) and engineering.
- Led remediation of high-risk security vulnerabilities alongside the cybersecurity team, identifying issues, driving fixes, and clearing all critical items within the required deadline.
- Built advanced web scraping and data-harvesting engines, ETL workflows, and automated file processing (JSON, CSV, XLSX) for large accounting datasets.
- Developed RESTful APIs (Flask, FastAPI) and automated report generators (Excel, CSV, PDF), reducing manual effort across corporate workflows.
- Owned AWS deployments (EC2, ECS, Lambda, load-balanced applications) and served as a primary PR reviewer, safeguarding code quality and delivery pace by architecting a robust QA workflow utilizing pytest, unittest, and Selenium for functional and end-to-end testing.

Start Future, São Paulo (part-time venture) — Co-Founder & Software Development Instructor

OCTOBER 2021 - JANUARY 2026

- Co-founded and co-led an education startup for 4+ years focused on affordable, career-ready tech training.
- Designed and ran an intensive 3-month bootcamp with four role-specific tracks (Back-end, Front-end, UX/UI, Scrum Master); taught and mentored 40+ students, with over 80% hired within months of completion.
- Took the company through FIAP's startup acceleration program, securing business partnerships and pitching to angel investors; placed top 10 in FIAP's annual student-startup competition.

Caixa Econômica Federal, São Paulo — Developer Intern

FEBRUARY 2018 - FEBRUARY 2020

- Started in high school at one of Brazil's largest public banks; moved from administrative work into development within 3 months, building VBA automations for spreadsheets, reporting, certificate generation, and direct-mail data processing.
- Took end-to-end ownership of several data pipelines by the second year, coordinating deliveries directly with other bank departments within a locked-down, intranet-only environment.

Education

FIAP, São Paulo — Postgraduate Degree / Master's (Lato Sensu Specialization), Data Engineering & Data Analytics

2024 - 2025

FIAP, São Paulo — Bachelor's Degree, Systems Analysis and Development

2021 - 2023

Volunteer Experience

Exotic India, via AIESEC, Delhi, India — Data Scientist (Computer Vision)

May 2022 - June 2022

- Built a Python/OpenCV application that automatically detected and cropped book covers and pages, reducing processing time from ~3 to ~1 minute per book (3x productivity gain) and shipped it as a standalone binary.
- Deployed the company's first data visualization environment (Grafana dashboards) and an MVP ELK stack with Docker for real-time data handling.

Awards

- **1st Place, FIAP "Improving the Planet with AI" Challenge:** built and pitched an environmental monitoring solution using computer vision.

- **1st Place, Daikin Enterprise Challenge:** designed the winning solution architecture and delivered the pitch.
- **2nd Place, Builtcode Enterprise Challenge:** recognized for top solution design.